computers

MDPI

Hypothesis

Reducing Peak Load Underprediction Through Risk-Aware Upper Quantile Forecasting: A University Laboratory Case Study

Marwa O. Al Enany [1,2], Mazen Hesham Elnahal [2] and Amira M. Gaber [1,2,*]

[1] Higher Institute of Computer Science and Information Systems, 6 October City 125-66, Giza, Egypt; marwa.osama@badyauni.edu.eg
[2] School of Computing and Data Science, Badya University, 6 October City 125-66, Giza, Egypt; m.hesham2400785@badyauni.edu.eg
* Correspondence: amira.mohamed@badyauni.edu.eg

Abstract

Underprediction of high electrical demand can be more operationally consequential than an equally sized overprediction, yet standard point forecasting models are optimized primarily for average error. This case study evaluates a multivariate quantile Transformer as a safety-oriented next observation forecasting layer for a university laboratory. A timestamp-level audit identified 33,374 native measurements collected from 22 April to 12 December 2024 at a median interval of approximately 10 min. The final leakage-free pipeline uses only real observations, performs the chronological split before sequence generation, fits all scalers on training data only, and rejects windows containing gaps greater than 30 min. Persistence, fixed-order SARIMA, LSTM, GRU, CNN–LSTM, and an MSE-trained Transformer were evaluated on the same 6595-sample test period. GRU achieved the best deterministic accuracy (MAE 0.017744 kW; RMSE 0.023278 kW), whereas the proposed $\tau = 0.90$ Transformer intentionally traded point accuracy (MAE 0.033830 $\pm$ 0.001133 kW) for asymmetric risk control. Across five independent runs, it achieved a pinball loss of 0.004453 $\pm$ 0.000069 kW, empirical coverage of 87.95 $\pm$ 1.01%, and a peak underprediction rate of 26.64 $\pm$ 5.32%, compared with 72.94–100% for the conventional benchmark outputs. Additional $\tau = 0.75$ and $\tau = 0.95$ experiments demonstrate the expected accuracy–safety trade-off. MAPE is not used as a primary metric because near-zero loads make percentage errors unstable. The results support the proposed model as a complementary upper quantile forecasting layer for this small, dynamic facility; they do not establish general performance at feeder or system scale.

Keywords: short-term load forecasting; quantile regression; transformer neural network; peak underprediction; empirical coverage; university laboratory; safety-oriented forecasting

Check for updates

Academic Editor: Wenbing Zhao

Received: 12 June 2026
Revised: 9 August 2026
Accepted: 10 August 2026
Published: 13 August 2026

Copyright: © 2026 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license.

1. Introduction

Accurate load forecasting is essential for smart grids, impacting energy management and balancing between supply and demand. Artificial Intelligence and machine learning have improved the prediction of complex electricity usage patterns. Short-term load forecasting (STLF) assists power companies in minimizing cost and enhancing operational decisions, hence enabling the integration of renewable energy sources through the management of their variation [1]. As urbanization and distributed energy resources (DERs) expand, accurate forecasting becomes crucial for grid stability. Hybrid deep learning models, such as Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), Convolutional

Neural Network (CNN), and Transformer architectures, have demonstrated efficacy in improving predictive performance by accurately capturing intricate load patterns [2].

Smart grids face operational and financial risks due to errors in power demand projections, especially underestimations. A 1% inaccuracy may result in financial losses for utilities amounting to 10 million currency units, compromising system safety and efficiency [3]. Accurate forecasts are crucial for maintaining supply–demand equilibrium, particularly with intermittent renewable energy sources (RESs), and are important for the efficient management of DERs and smart meters [4]. Neglecting this might result in ineffective asset management, impacting grid resilience and energy equity. Consequently, reducing prediction errors is essential for economic stability and a dependable electricity supply.

Time series energy forecasting, especially in smart grids, faces several challenges due to sophisticated consumption patterns, varied energy sources, and the requirement for accuracy [5]. Conventional statistical techniques are inadequate for addressing non-linear trends and seasonal variations, whereas the uncertainty from renewable sources and local demand shifts need precise predictions. Data management is essential because of the large amount of data generated by smart meters, as data quality influences the efficacy of machine learning methods. Modern CNN techniques are insufficient for feature extraction, requiring the development of novel architectures such as 1D-UCNN [6] and Bi-LSTM [7]. Real-time applications need efficiency and must resolve privacy issues, possibly via split learning. Progress in AI and machine learning must concentrate on hybrid models and optimization to improve predictive accuracy in smart grids.

Undershooting Bias in Power Consumption is the consistent bias of forecasting methods to underestimate peak electricity demand compared to the actual peak demand. This directional error arises notably under extreme conditions, such as heatwaves or cold snaps, since predictions underestimate the maximum load [8]. The bias originates from the underlying model's architecture or training, leading to significant underestimations of peak consumption. Consequently, projections driven by this bias may produce an excessively positive assessment of the system's operating capacity, presenting scenarios that are less severe than the real circumstances, thus compromising grid reliability.

The inability of conventional models to accurately forecast peak demand arises not from arbitrary errors, but from underlying limitations in their design and intent. This issue can be understood through three interrelated perspectives: statistical, behavioral/physical, and economic.

Statistical perspective: Conventional techniques such as time series studies and machine learning regressions try to reduce average error, resulting in suboptimal performance during anomalous events [9]. These models have difficulty predicting exceptional peaks because they concentrate on standard conditions, leading to conservative projections in the presence of unexpectedly extreme inputs, which results in an undershooting bias and restricted extrapolation abilities.

Behavioral/physical perspective: Electrical demand depends nonlinearly on temperature and other exogenous drivers, and flexible load-forecasting models commonly represent temperature effects using nonlinear or nonparametric terms [10]. Demand peaks are also influenced by factors such as humidity, time, and user behavior, which conventional models may inadequately capture. Furthermore, conventional models often neglectthe physical constraints of demand-side infrastructure, presenting load as continuous rather than as a strained system, leading to unexpected demand plateaus and outages.

Economic perspective: Demand forecasting models encounter numerous problems. For instance, they neglect peak instances, resulting in a bias toward non-extreme events.

They lack real-time stress indicators, such as historical peak data and evolving market trends that influence consumer behavior.

This study evaluates a multivariate Transformer trained using quantile regression to reduce directional underprediction risk in a small, dynamic university laboratory load. The revised analysis distinguishes point accuracy from safety-oriented upper quantile forecasting and avoids presenting these as interchangeable objectives. The final experiment is based on a timestamp-level audit and a leakage-free native observation task rather than on synthetic 3 min resampling. Its main contributions are: (1) a causal preprocessing pipeline that uses only real observations and fits transformations on the training period; (2) a fair same-period benchmark suite comprising persistence, SARIMA, LSTM, GRU, CNN–LSTM, and an MSE Transformer; (3) a multivariate $\tau = 0.90$ Transformer evaluated with pinball loss, empirical coverage, calibration error, violation rate, and peak subset metrics; (4) a multi-quantile analysis at $\tau = 0.75$, 0.90, and 0.95; and (5) native data ablation and permutation feature importance that separate the effects of electrical inputs, asymmetric loss, and architectural depth. The findings are framed explicitly as a single-site case study.

2. Related Work

Most existing power consumption forecasting methods, including traditional statistical models, such as Autoregressive Integrated Moving Average (ARIMA) and Seasonal ARIMA (SARIMA), face limitations due to their structural designs and loss function choices. While these traditional methods are effective in capturing linear trends and periodic features, such as in predicting electric vehicle power demand, they may not adequately address the complexities of modern forecasting needs [11]. The authors in [12] employed ARIMA modeling to analyze energy consumption trends of laboratory equipment over a month, emphasizing improved energy management. Despite the model's limitations in incorporating essential features, it was effectively applied to laboratory equipment, focusing purely on past consumption data. The authors in [13] apply a seasonal ARIMA model for a distinct medium-term (30-month) forecast, trying to predict "critical power days" to enhance grid performance and control adverse conditions. In contrast, authors in [14] concentrate on a short-term perspective, employing non-seasonal as well as seasonal sliding-window ARIMA algorithms to predict hourly load at the district meter level, while incorporating an online approach for dynamic updates. Authors in [15] clearly apply the integration component of the ARIMA model to address non-stationarity in smart meter data; nevertheless, their primary contribution is in incorporating this 12-week forecast into an intelligent clustering method to develop a dynamic demand response model. A separate study [16] emphasizes model flexibility for short-term forecasting in distribution systems, contrasting both frequentist and Bayesian methodologies for parameter estimation in univariate multiplicative seasonal ARIMA models.

By contrast, some other previous studies implemented deep learning models, particularly LSTM and GRU networks, to enhance forecasting by identifying non-linear patterns and long-term dependencies in sequential data. One of these studies [17] presents a comprehensive investigation of deep learning developments, identifying LSTM as a fundamental solution for continuous time series analysis. This fundamental utility is illustrated in practical applications by [18], which utilizes LSTM for CO_2 emissions forecasting, and [19], which demonstrates its appropriateness for IoT analytics in dynamic contexts, both reflecting its mechanisms for mitigating gradient-related issues. Authors in [20] propose modifications to the forgetting mechanism to achieve slower decay and improved long-range modeling in complex networks. Consequently, although LSTMs incorporate gating mechanisms that substantially eliminate the vanishing gradient issue and improve long-term information retention, studies argue that their basic forget mechanism may still result in exponential

information decay, requiring ongoing modifications for the optimal modeling of complex systems across extensive sequences.

Advancements in the use of sophisticated neural models like GRU are demonstrated by research such as [21], which attempts to enhance the fundamental GRU architecture via their Drop-GRU variant to improve efficiency and normalization for consumption forecasting. Expanding on the evaluation of core models, another study [2] takes a comprehensive, ensemble-oriented approach, methodically testing hybrid combinations of the model types (GRU, CNN, LSTM), integrating them with traditional ML regressors to discover optimal predictive ensembles. An integrated hybrid-centric design that combines Transformer, TCN, and GRU components to effectively capture complex temporal relationships in campus heat loads is presented in [22]. Nonetheless, recurrent networks such as LSTMs and GRUs proved to have limitations involving huge processing demands, the potential for vanishing or exploding gradients in deep designs, and constraints in modeling sophisticated multi-scale temporal patterns. Their recurrent architectures may obstruct parallel processing, resulting in longer training periods and sensitivity to hyperparameter tuning.

In contrast, Transformer models utilize self-attention mechanisms for time series forecasting, improving the management of long-range dependencies in data like power consumption, compared to recurrent networks. For instance, The work presented in [23] proposed a self-attention Transformer-based deep learning framework that estimates energy consumption in commercial buildings, integrating historical energy use, meteorological conditions, and occupancy data and resulting in a 23.7% enhancement in prediction accuracy. Combining a Temporal Fusion Transformer (TFT) with CNN, The work presented in [24] proposed the HTFT-CNN model with an attention mechanism for improved time series analysis and feature integration, producing reduced Root Mean Square Error (RMSE) and mean absolute percent error (MAPE) metrics. A different hybrid short-term load forecasting (STLF) model for smart grids [25], combining K-shape time series clustering with a Transformer neural network, achieved a 2% to 5% reduction in MAPE.

As power systems expand with growing incorporation of renewable energy and smart grids, the necessity for reliable forecasting models arises [26]. It focuses on creating an accurate power load forecasting model for residential smart grids by integrating LSTM and Transformer architectures for multi-step forecasting and predicting total power load over several future time steps. A Transformer-based model designed for interpretable short-term forecasting of residential net load is proposed in [27] with a Generative Transformer Prediction Module that employs quantile regression for multi-step forecasting. The model outperforms benchmarks such as LSTM and Vanilla Transformer in accuracy according to their results.

Although most studies improve performance metrics and increase the accuracy of forecasting, most of them have limitations and challenges; as shown in [23], there were challenges regarding long-term adaptation, generalizability for extensive datasets, and the incorporation of renewable energy, indicating potential avenues for future investigation. The work presented in [24] is hindered by high computational costs and reliance on clean historical data. The model in [25] faces challenges such as data dependency, trade-offs in interpretability, and high computational complexity. Also, [27] faces limitations including a focus on single households, challenges in deep learning interpretation, high computational demands, and reliance on data quality.

Recent proposals include a hybrid method combining SARIMA and deep learning for EV predictions and an integrated system like LITE-FORT [28] that employs ARIMA, CNN, and LSTM for user consumption forecasting. The Adaptformer [29] seeks to improve long-term forecasting by addressing the intricacies of high-dimensional datasets, emphasizing

the necessity for adaptive models capable of managing the dynamic challenges posed by modern energy systems and the associated costs of varied forecasting errors.

Regarding the impact of loss functions in time series forecasting, the comparative study in [30] summarizes 14 regression loss functions for time series forecasting, highlighting their optimal applications for enhancing model convergence. Key functions include mean absolute error (MAE) and root mean squared error (RMSE), which provide scalable solutions, concluding that Quantile Loss is beneficial for generating prediction intervals in various models. In [31], authors enhanced electricity demand forecasting using an LSTM neural network, employing two novel loss functions: Weighted Mean Squared Error (WMSE) to emphasize high-demand periods and Negative Log Likelihood (NLL) to capture prediction uncertainty. The results indicate that LSTM models achieved a MAPE of $5.54 \pm 1.00\%$.

A complementary perspective is provided by Stergiou and Karakasidis [32], who combined recurrent deep learning models with chaos theory analysis and a Lyapunov time interpretation for Greek load forecasting. Their results emphasize that model accuracy must be interpreted together with the forecast horizon and the dynamical predictability of the underlying series. The present study does not estimate a Lyapunov exponent; instead, it adopts a conservative one-step, next native observation task and reports the actual horizon characteristics of the accepted sequences.

3. Methodology

The final study design forecasts the next real laboratory measurement from the previous 20 real observations. It does not construct synthetic intermediate targets and does not claim recovery of events occurring between native measurements. The methodological workflow combines a timestamp audit, chronological splitting, leakage-free sequence construction, fair same-input benchmarks, and quantile-specific evaluation. The earlier two-phase prototype remains documented as development context, but all final numerical claims come from the native observation experiment described below.

3.1. Development Context and Final Experimental Design

The initial Phase I prototype used a univariate Transformer, a long historical input, last-observation carry-forward for missing values, and a peak-weighted objective. It was useful for identifying undershooting behavior but combined several changes and did not provide a fair basis for attributing performance gains. Figure 1 is retained to document that historical developmental path; the prototype is not used as the principal comparator in the final benchmark tables. The historical Phase I workflow is summarized in Figure 1.

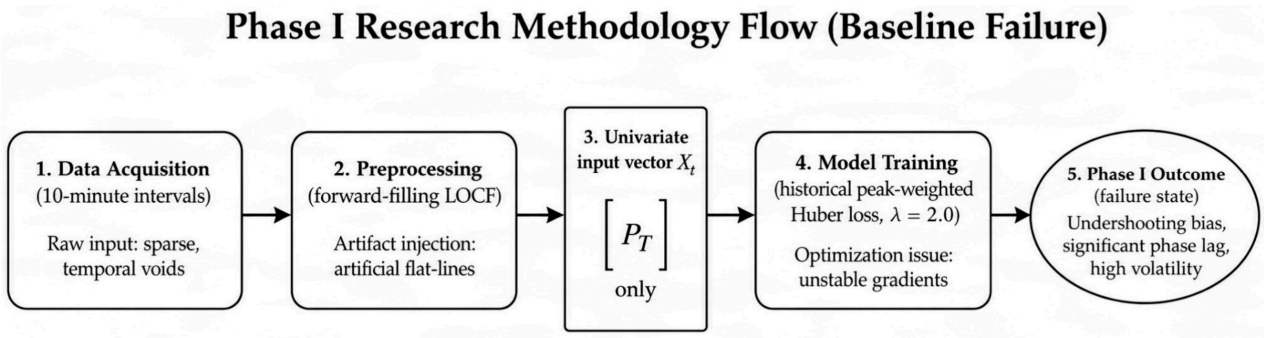


Figure 1. Historical Phase I development workflow (baseline failure mode).

The final experimental design was rebuilt after auditing the source timestamps. All primary models now receive comparable native observation sequences and are evaluated on the same chronological test period. The proposed model differs from the deterministic

neural benchmarks mainly in its pinball loss objective and upper quantile interpretation. The leakage-free final workflow is summarized in Figure 2.

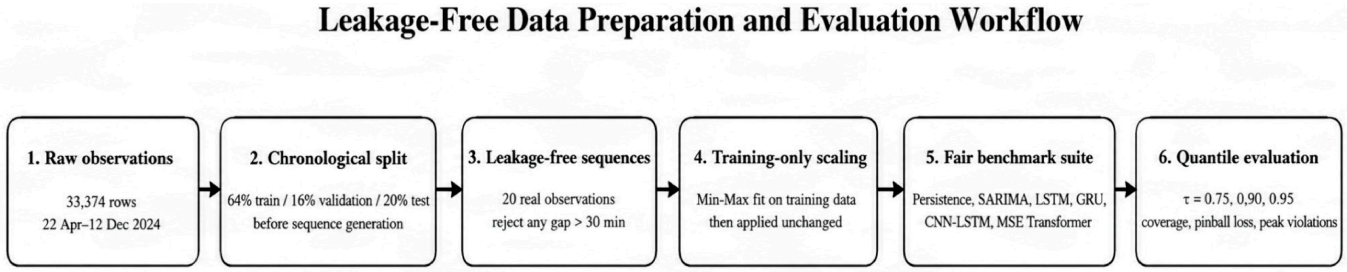


Figure 2. Leakage-free data preparation and evaluation workflow used for the final experiments.

3.2. Dataset Description and Timestamp Audit

The open electrical measurement dataset was collected from a university laboratory and is available through the Version v1 10.5281/zenodo.17107426 repository cited in [33]. The source files contain total active power, three-phase currents, voltages, power factor, frequency, and related quantities. The present forecasting task uses total active power P_T, currents I_A, I_B, and I_C, and two cyclical time-of-day features. Because the measurements describe one laboratory rather than a feeder or regional grid, the analysis is presented as a facility-scale case study. The six forecasting variables are summarized in Table 1.

Table 1. Variables used by the forecasting models.

Feature	Symbol	Role	Description and Rationale
Total active power	P_T	Input and target	Measured laboratory active power in kW; the next real observation is forecast.
Phase A current	I_A	Input	Three-phase physical covariate; permutation analysis shows the strongest auxiliary contribution.
Phase B current	I_B	Input	Complements Phase A and captures phase-specific load distribution.
Phase C current	I_C	Input	Completes the three-phase representation; contribution is smaller in this dataset.
Hour sine	$hour_{sin}$	Input	$\sin(2\pi h/24)$, preserving time-of-day cyclicity.
Hour cosine	$hour_{cos}$	Input	$\cos(2\pi h/24)$, paired with $hour_{sin}$ to avoid a midnight discontinuity.

After concatenation, timestamp parsing, chronological sorting, and removal of one duplicate timestamp, the verified dataset contains 33,374 complete rows from 22 April 2024 17:01 to 12 December 2024 10:29. The median native interval is 9.9999 min. All months from April through December are represented; there are no completely missing months. Relative to a regular grid at the median interval, approximately 0.84% of expected timestamps are absent. The audit identified 154 gaps longer than 15 min and four gaps longer than 2 h, motivating explicit sequence rejection rather than interpolation [34]. The verified audit, chronological partitions, and peak definition are summarized in Table 2.

Table 2. Verified data audit, chronological partition, and peak definition.

Item	Verified Value
Usable native observations	33,374
Observation period	22 April–12 December 2024
Median native interval	Approximately 10 min
Completely missing months	None
Input sequence	Previous 20 real observations; typically about 190 min from first to last input
Forecast target	Next real observation; median horizon approximately 10 min
Maximum allowed consecutive gap	30 min, including the final input-to-target transition
Training sequences	21,015 (64%)
Validation sequences	5259 (16%)
Test sequences	6595 (20%)
Peak definition	Training target 90th percentile, $P_T \geq 0.154$ kW
Peak test samples	521

3.3. Leakage-Free Preprocessing and Sequence Construction

3.3.1. Data Ingestion and Chronological Partitioning

Column names were standardized, the timestamp column was parsed to a date time index, and files were concatenated and sorted. The raw observed rows were divided chronologically into 64% training, 16% validation, and 20% testing before sequence construction or any transformation. This ordering prevents information from the validation or test periods from influencing training time preprocessing.

3.3.2. Missing Data and Gap-Aware Sequence Rejection

No resampling, linear interpolation, forward filling, or synthetic intermediate timestamp is used in the primary analysis. Each candidate sequence contains 20 actual observations, and its target is the next actual observation [35]. A sequence is accepted only when every consecutive timestamp difference, including the final transition to the target, is at most $\Delta_{max} = 30$ min. For a candidate sequence with timestamps $t_0, \ldots, t_T$, the validity indicator is

$$M_{seq} = \prod_\{k = 1\}\hat{}\{T\} \mathbb{1}(t_k - t_{k-1} \leq \Delta_{max}). \quad (1)$$

Only sequences with $M_{seq} = 1$ are retained. This rule removes long outages while preserving ordinary timing jitter in the native approximately 10 min measurements. Every accepted target is a real observation, and every input source timestamp precedes its target [36].

3.3.3. Feature Engineering and Scaling

The hour of day h is represented by two cyclical variables [37]:

$$hour_{sin} = \sin(2\pi h/24), \quad hour_{cos} = \cos(2\pi h/24). \quad (2)$$

The resulting feature vector is

$$X_t = [P_{T,t}, I_{A,t}, I_{B,t}, I_{C,t}, hour_{sin,t}, hour_{cos,t}]. \quad (3)$$

Separate MinMaxScaler instances are fitted exclusively on complete training rows for the six inputs and the target. The fitted scalers are then applied without refitting to validation and test sequences. All metrics are calculated after inverse transformation to kW [38].

3.4. Forecasting Models

3.4.1. Historical Baseline Transformer

The earlier developmental baseline used a univariate Transformer based only on P_T and is retained for methodological continuity in Figure 3. Because it used a different input definition and preprocessing design, it is not used for the final headline comparison. The final fairness analysis instead uses the benchmark suite in Table 3.

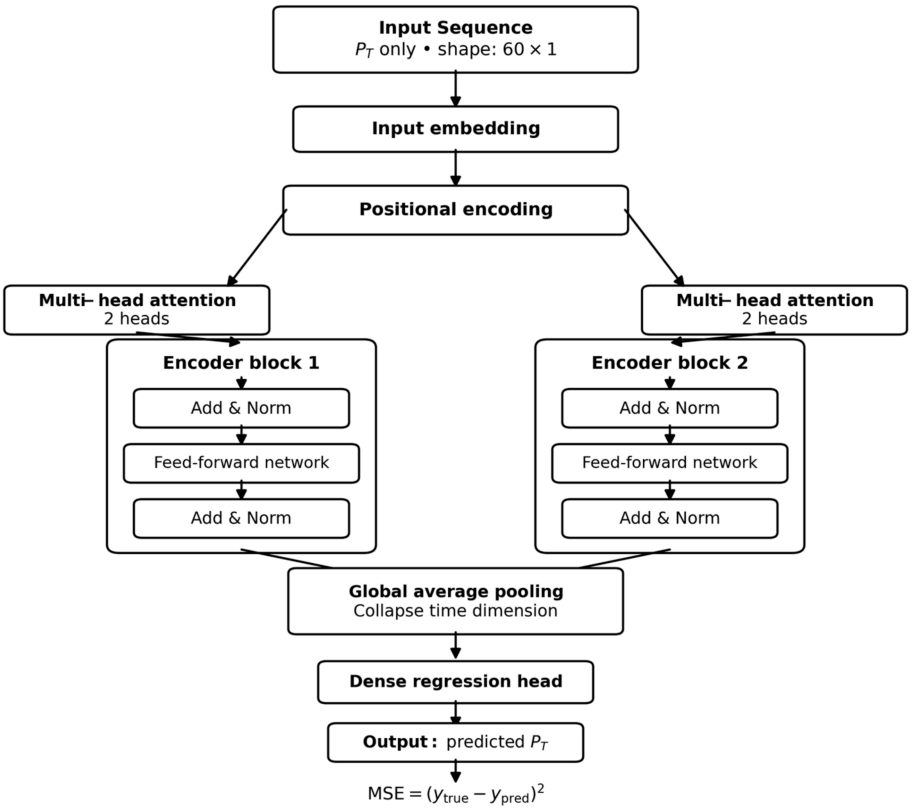


Figure 3. Historical univariate Transformer architecture from the initial development phase.

Table 3. Models included in the final same-period comparison.

Model	Input	Objective	Purpose
Persistence	Latest P_T	None	Naïve sanity check baseline
SARIMA(1,0,0) × $(1,0,0)_{144}$	P_T	Conditional sum of squares	Seasonal statistical reference
LSTM	20 × 6	MSE	Recurrent benchmark
GRU	20 × 6	MSE	Compact recurrent benchmark
CNN–LSTM	20 × 6	MSE	Local pattern plus recurrent benchmark
Transformer (MSE)	20 × 6	MSE	Architecture-controlled deterministic comparator
Proposed Transformer	20 × 6	Pinball, $\tau = 0.90$	Upper quantile safety-oriented forecast

3.4.2. Fair Benchmark Suite

Persistence predicts the next P_T value as the most recent observed P_T. A fixed multiplicative SARIMA(1,0,0) × $(1,0,0)_{144}$ model represents daily seasonality at the approximately 10 min native interval. The LSTM, GRU, CNN–LSTM, and MSE Transformer receive the same 20 × 6 multivariate input and chronological partitions as the proposed model. The classical model is inherently univariate. Deterministic neural comparators use MSE, whereas the proposed Transformer uses pinball loss. The same-period benchmark suite is summarized in Table 3.

3.4.3. Proposed Multivariate Quantile Transformer

The proposed model projects the six input variables to a 64-dimensional representation and adds fixed sinusoidal positional encoding. Two Transformer encoder blocks are used. Each block contains four self-attention heads with a per-head dimension of 16, residual connections, layer normalization, dropout, and a feed-forward dimension of 32. A learned attention pooling layer aggregates the temporal representation, followed by a 64-unit ReLU multilayer perceptron and a single linear output. The final architecture contains 46,914 trainable parameters. The proposed architecture is summarized in Figure 4, and its exact training configuration is listed in Table 4.

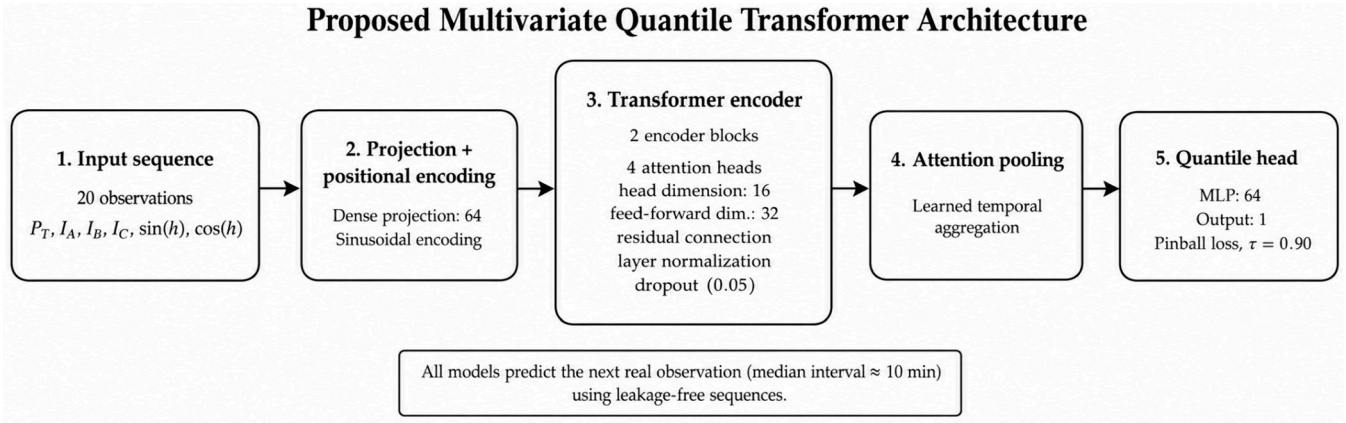


Figure 4. Proposed multivariate quantile Transformer architecture.

Table 4. Final proposed model and training configuration.

Hyperparameter	Value
Input features	P_T, I_A, I_B, I_C, $hour_{sin}$, $hour_{cos}$
Sequence length	20 real observations
Forecast horizon	Next real observation; median approximately 10 min
Projection/model dimension	64
Encoder blocks	2
Attention heads	4
Per-head dimension	16
Feed-forward dimension	32
Attention pooling	Learned temporal aggregation
MLP hidden units	64, ReLU
Dropout	0.05

Table 4. *Cont.*

Hyperparameter	Value
Positional encoding	Fixed sinusoidal
Optimizer	Adam
Learning rate	0.0008
Gradient clipping	Global norm 1.0
Batch size	128
Maximum epochs	35
Early-stopping patience	4
ReduceLROnPlateau patience	2
Loss	Pinball loss, $\tau = 0.90$
Independent proposed model runs	5 seeds: 11, 23, 37, 53, 71

3.5. Loss Functions

3.5.1. Historical Peak-Weighted Huber Loss

For completeness, the historical peak-weighted Huber formulation used during the exploratory development phase is stated below. Let $e = y - \hat{y}$. The Huber component is

$$H_\delta(e) = \{½e^2, \text{ if } |e| \leq \delta; \delta(|e| - ½\delta), \text{ if } |e| > \delta\}. \quad (4)$$

The peak-weighted historical loss is therefore

$$L(y, \hat{y}) = \{\lambda H_\delta(y - \hat{y}), \text{ if } y > \theta \text{ and } \hat{y} < y; H_\delta(y - \hat{y}), \text{ otherwise}\}. \quad (5)$$

Here λ is the additional underprediction penalty and θ is a peak threshold. This exploratory objective is documented only to correct the former mathematical inconsistency; it is not the objective used for the final proposed model or benchmark claims.

3.5.2. Quantile Regression and Pinball Loss

For target quantile τ, the final model minimizes

$$L_\tau(y, \hat{y}_\tau) = \max[\tau(y - \hat{y}_\tau), (\tau - 1)(y - \hat{y}_\tau)]. \quad (6)$$

When $y > \hat{y}_\tau$, the observation exceeds the predicted upper bound and the error receives weight τ. When $y \leq \hat{y}_\tau$, the prediction is conservative and receives weight $1 - \tau$. At $\tau = 0.90$, underprediction is therefore weighted nine times more strongly than overprediction of the same magnitude. The output is a conditional upper quantile, not a formal physical guarantee. Because only one upper quantile is produced in the principal model, prediction interval width is not applicable.

3.6. Training Procedure and Reproducibility

Models were implemented in TensorFlow/Keras and trained on an NVIDIA T4 GPU using mixed precision. Training, validation, and test arrays remained chronological. Early stopping monitored validation loss and restored the best weights; ReduceLROnPlateau adjusted the learning rate when validation loss plateaued, and gradient clipping was used for stability [39]. The proposed $\tau = 0.90$ model was trained with five independent random seeds and is reported as mean $\pm$ standard deviation. The other neural benchmarks, ablation configurations, and $\tau = 0.75/0.95$ sensitivity models use one fixed seed each, which is stated

wherever results are shown. The fixed-order SARIMA baseline uses conditional sum of squares estimation as a reproducible, memory-stable statistical reference.

3.7. Evaluation Metrics

All errors are evaluated after inverse scaling to kW. Mean absolute error and root mean squared error are

$$\mathrm{MAE} = (1/n)\sum |\, y_i - \hat{y}_i \,|, \qquad \mathrm{RMSE} = \sqrt{[(1/n)\sum (y_i - \hat{y}_i)^2]}. \tag{7}$$

R^2 summarizes explained variance. Mean error is defined as mean($\hat{y} - y$), so positive values denote conservative overprediction and negative values denote underprediction. Peaks are test observations at or above the training-target 90th-percentile threshold of 0.154 kW. Peak MAE, peak mean error, maximum absolute peak error, and peak violation rate are reported. MAPE is excluded from primary evaluation because zero and near-zero actual loads make its denominator unstable and its values potentially misleading.

Empirical coverage for quantile τ is

$$\mathrm{Coverage}_\tau = (1/n)\sum \mathbb{1}(y_i \le \hat{y}_{\tau,i}). \tag{8}$$

Calibration error is $|\mathrm{Coverage}_\tau - \tau| \times 100$ percentage points, and violation rate is $100 \times (1 - \mathrm{Coverage}_\tau)$. Test pinball loss is the sample mean of Equation (6). These standard terms replace the earlier ambiguous label "Safety Bias."

4. Evaluation and Results

4.1. Experimental Environment and Data Verification

The final notebook used Python 3.12.13 (Python Software Foundation, Wilmington, DE, USA), TensorFlow 2.20.0 and Keras 3.13.2 (Google LLC, Mountain View, CA, USA), pandas 2.2.2, NumPy 2.0.2, scikit-learn 1.6.1, and an NVIDIA T4 GPU (NVIDIA Corporation, Santa Clara, CA, USA). The complete leakage audit passed: raw observations were split before sequence creation, no bidirectional interpolation was used, every target was a real measurement, every input source timestamp preceded its target, scalers were fitted on training data only, and test counts agreed across benchmark and calibration analyses. The principal test set contains 6595 sequences, including 521 peak targets.

4.2. Same-Input Deterministic Benchmark Comparison

GRU achieved the lowest MAE and RMSE, closely followed by CNN–LSTM. The proposed Transformer does not outperform the deterministic neural models on point accuracy because its objective estimates an upper conditional quantile rather than the conditional mean. The manuscript therefore no longer presents the proposed model as the universally most accurate point forecaster. The deterministic benchmark results are summarized in Table 5.

Table 5. Same-period deterministic benchmark results on 6595 test sequences.

Model	MAE (kW)	RMSE (kW)	R^2
GRU	0.017744	0.023278	0.7724
CNN–LSTM	0.017816	0.023360	0.7708
LSTM	0.018741	0.024404	0.7499
Transformer (MSE)	0.018812	0.024428	0.7494

Table 5. *Cont.*

Model	MAE (kW)	RMSE (kW)	R^2
Persistence	0.024744	0.033692	0.5232
Proposed Transformer, $\tau = 0.90$	0.033830 ± 0.001133	0.041229 ± 0.001207	0.2855 ± 0.0422
SARIMA(1,0,0) $\times$ $(1,0,0)_{144}$	0.038675	0.049346	-0.0228

The proposed model is mean $\pm$ SD across five independent runs; other learned benchmarks use one fixed seed. Lower MAE and RMSE are better.

4.3. Peak and Quantile-Oriented Performance

Across five runs, the proposed model achieved $87.95 \pm 1.01\%$ overall coverage against the 90% target, corresponding to a calibration error of 2.05 ± 1.01 percentage points. Its main operational advantage appears during peaks: the peak violation rate falls to $26.64 \pm 5.32\%$, whereas the comparator outputs underpredict 72.94–100% of peak observations. This improvement is obtained through a positive mean peak error and should be interpreted as a deliberate safety margin rather than a free accuracy gain. Figures 5–7 illustrate representative high-load behavior, error distributions, and predicted-versus-actual relationships, while Table 6 summarizes the quantile-oriented and peak-subset metrics. Figures 8 and 9 further show time-series residuals and absolute error across load bins.

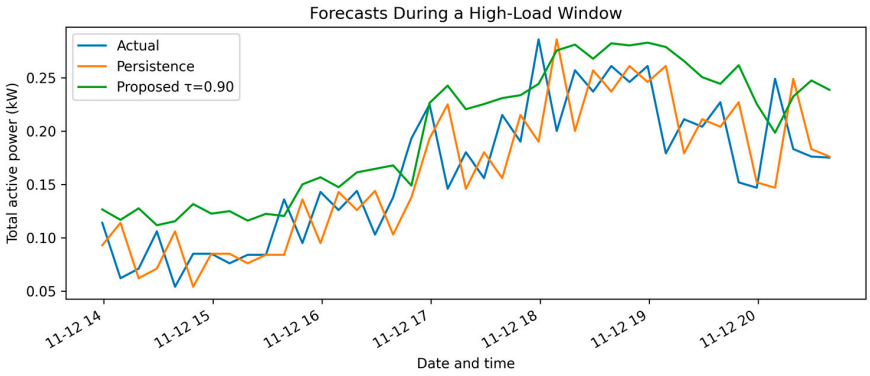


Figure 5. Example high-load test window comparing actual load, persistence, and the proposed $\tau = 0.90$ forecast.

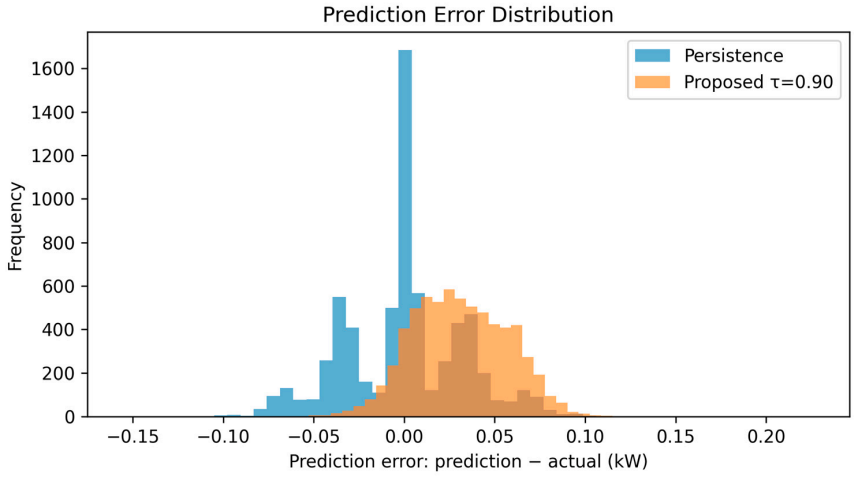


Figure 6. Distribution of prediction errors for the final same-period benchmark outputs.

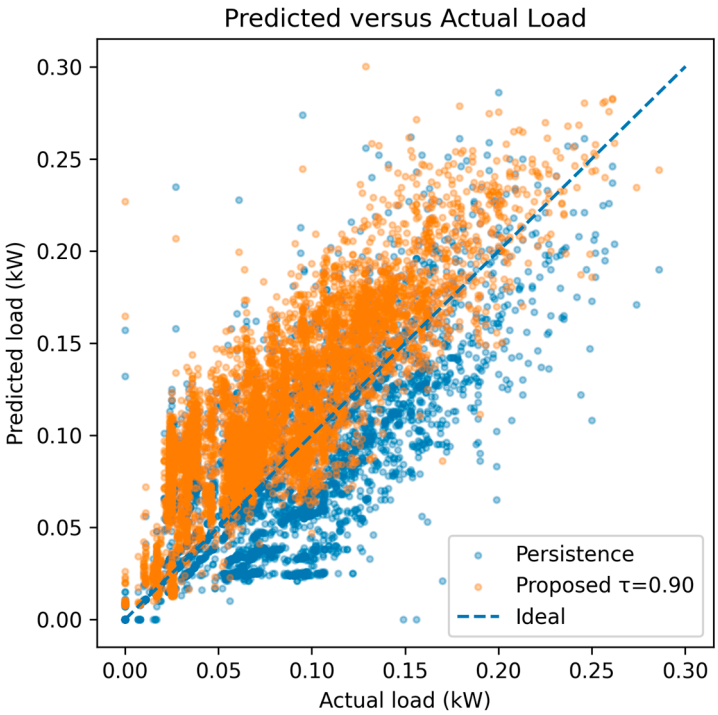


Figure 7. Predicted versus actual load for selected benchmark models on the leakage-free test period.

Table 6. Quantile-oriented and peak subset comparison at $\tau = 0.90$.

Model	Pinball Loss (kW)	Coverage (%)	Overall Violation (%)	Peak MAE (kW)	Peak Mean Error (kW)	Peak Violation (%)
Persistence	0.012355	59.17	40.83	0.03710	−0.02336	72.94
LSTM	0.008486	55.72	44.28	0.02842	−0.02092	77.35
GRU	0.007860	58.10	41.90	0.02629	−0.01874	76.39
CNN–LSTM	0.008156	56.54	43.46	0.02588	−0.01884	76.78
Transformer (MSE)	0.009077	52.80	47.20	0.02708	−0.01952	76.97
Proposed $\tau = 0.90$	0.004453 ± 0.000069	87.95 ± 1.01	12.05 ± 1.01	0.02691 ± 0.00290	+0.01775 ± 0.00469	26.64 ± 5.32
SARIMA	0.022265	52.98	47.02	0.11377	−0.11377	100.00

Positive peak mean error indicates conservative overprediction. The proposed model is mean ± SD across five runs.

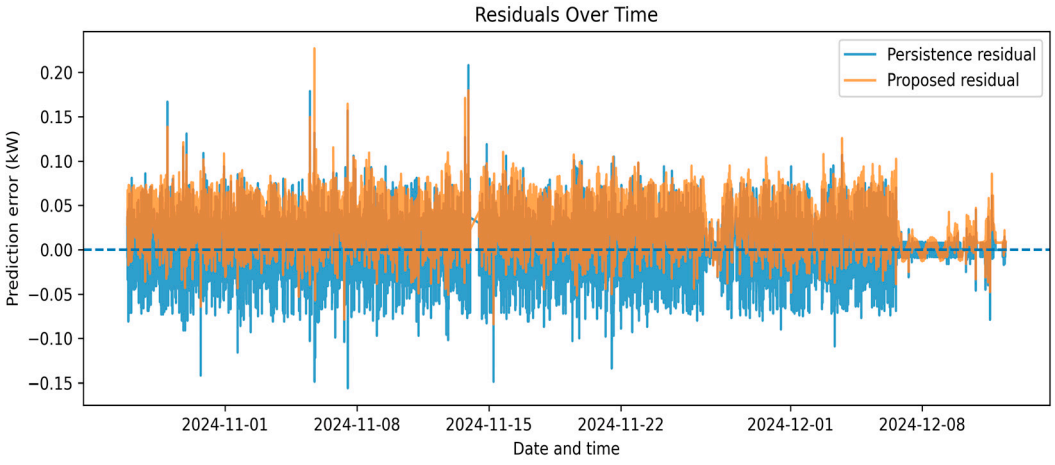


Figure 8. Time series residuals on the final leakage-free test period.

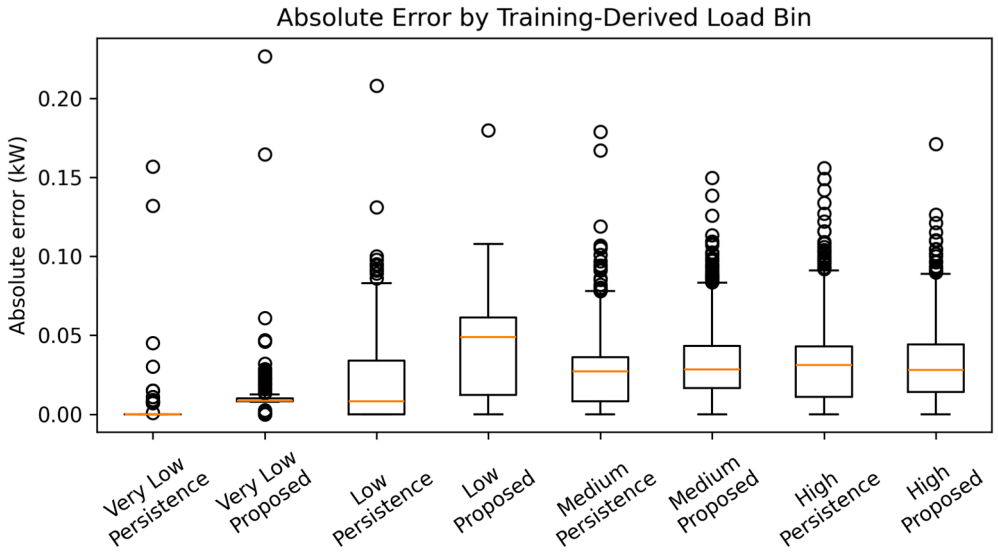


Figure 9. Absolute prediction error by actual load quartile.

4.4. Calibration by Load Level

Calibration is not uniform across load levels. The proposed model is close to the 90% target in the low and medium bins, overcovers the small very-low-load bin, and undercovers the high-load bin at 82.03%. The high-load value remains materially higher than the 38.49% coverage of persistence and the MSE Transformer, but it confirms that the model is not a formal safety guarantee and may benefit from load-conditional recalibration. The load-stratified calibration results are reported in Table 7.

Table 7. Diagnostic $\tau = 0.90$ calibration by actual load quartile.

Load Bin	N	P_T Range (kW)	Persistence Coverage (%)	MSE Transformer Coverage (%)	Proposed Coverage (%)	Proposed Pinball (kW)
Very low	167	<0.011	95.81	78.44	100.00	0.001322
Low	1882	0.011–0.044	88.68	83.69	89.00	0.004452
Medium	2782	0.044–0.102	50.11	39.43	90.94	0.003956
High	1764	≥0.102	38.49	38.49	82.03	0.005316

The proposed values use the mean prediction of the five $\tau = 0.90$ runs for load-bin diagnostics.

4.5. Multi-Quantile Accuracy–Safety Trade-Off

The additional quantiles provide the requested accuracy–safety trade-off. The $\tau = 0.75$ model has the lowest MAE among the quantile variants but permits more violations. The $\tau = 0.95$ model provides the strongest safety margin and lowest pinball loss at its own target, but its point error is substantially larger. The $\tau = 0.90$ setting is therefore an intermediate operating point rather than a universally optimal choice. Table 8 summarizes the operating-point comparison, and Figures 10 and 11 visualize calibration and the accuracy–underprediction trade-off.

Table 8. Quantile operating point comparison.

τ	MAE (kW)	RMSE (kW)	Pinball Loss (kW)	Coverage (%)	Overall Violation (%)	Peak Coverage (%)	Peak Violation (%)
0.75	0.023666	0.030331	0.007741	74.12	25.88	51.44	48.56

Table 8. *Cont.*

τ	MAE (kW)	RMSE (kW)	Pinball Loss (kW)	Coverage (%)	Overall Violation (%)	Peak Coverage (%)	Peak Violation (%)
0.90	0.033830 ± 0.001133	0.041229 ± 0.001207	0.004453 ± 0.000069	87.95 ± 1.01	12.05 ± 1.01	73.36 ± 5.32	26.64 ± 5.32
0.95	0.054531	0.060996	0.003061	96.85	3.15	89.06	10.94

τ = 0.90 is mean ± SD across five runs; τ = 0.75 and 0.95 are fixed seed sensitivity experiments.

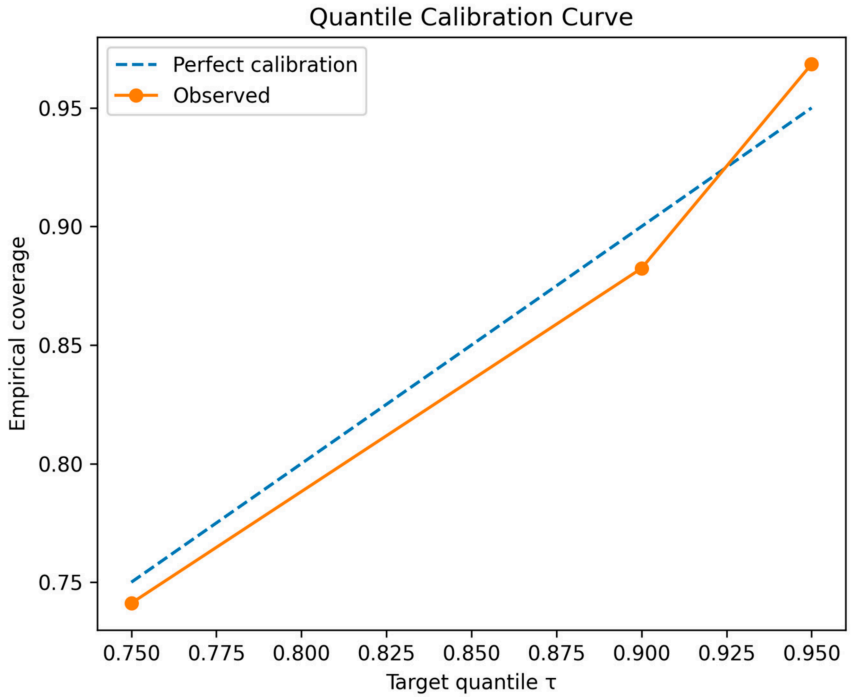


Figure 10. Empirical coverage at τ = 0.75, 0.90, and 0.95 compared with ideal calibration.

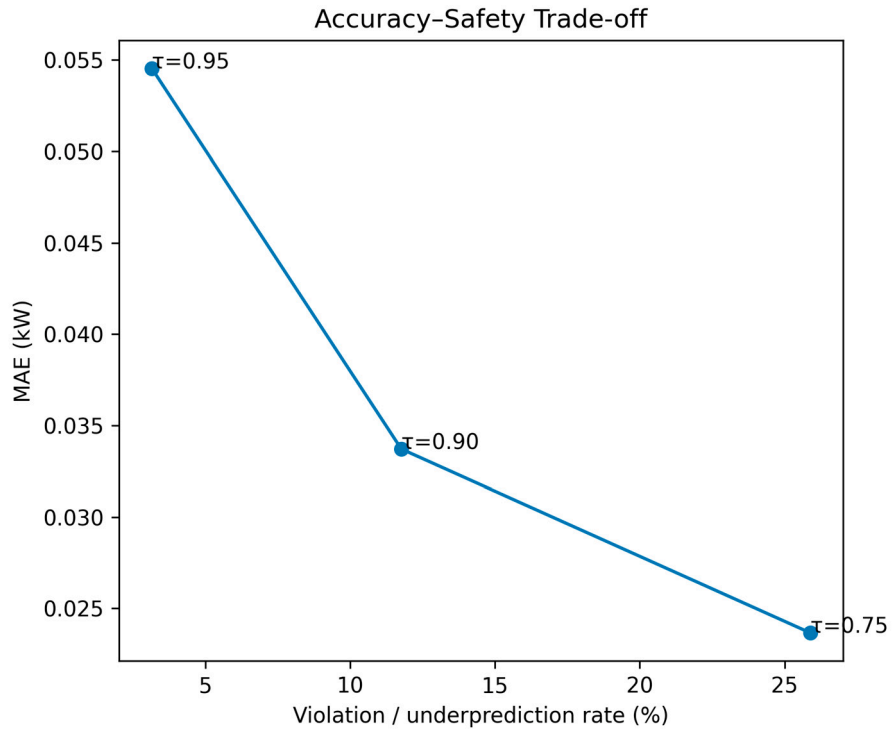


Figure 11. Point accuracy and underprediction trade-off across quantile operating points.

4.6. Native Data Ablation Study

The timestamp audit showed that the source instrumentation is natively sampled at approximately 10-min intervals. Consequently, a leakage-free 3 min experiment cannot create independent physical measurements or support transient recovery claims. The final ablation therefore uses native observations and isolates input features, loss function, and architectural depth. The native-data ablation results are summarized in Table 9.

Table 9. Leakage-free native observation ablation (one fixed seed per configuration).

Configuration	Inputs	Loss	Blocks	MAE (kW)	RMSE (kW)	Coverage (%)	Peak Mean Error (kW)	Peak MAE (kW)
N0	P_T only	MSE	2	0.019896	0.026064	53.01	−0.022052	0.029404
N1	6 features	MSE	2	0.018812	0.024428	52.80	−0.019519	0.027082
N2	6 features	Pinball 0.90	2	0.033423	0.040508	86.81	+0.010498	0.022293
N3	6 features	Pinball 0.90	4	0.035860	0.043232	89.25	+0.021933	0.029222

Adding currents and time features to the MSE Transformer reduces MAE from 0.019896 to 0.018812 kW and makes the negative peak bias less severe. Replacing MSE with $\tau = 0.90$ pinball loss reverses the peak mean error from negative to positive and reduces peak MAE, but increases overall MAE and RMSE. Increasing depth improves coverage toward the 90% target while further increasing point error. Thus, multivariate inputs mainly provide a modest accuracy gain, whereas pinball loss drives the directional safety behavior.

4.7. Permutation Feature Importance

Recent active power remains the dominant predictor. Phase A and Phase B currents nevertheless provide measurable complementary information, supporting their inclusion without claiming that current signals dominate the forecast. The cyclical variables and Phase C current have smaller effects in this dataset. Figure 12 visualizes the permutation ranking, and the numerical importance values are reported in Table 10.

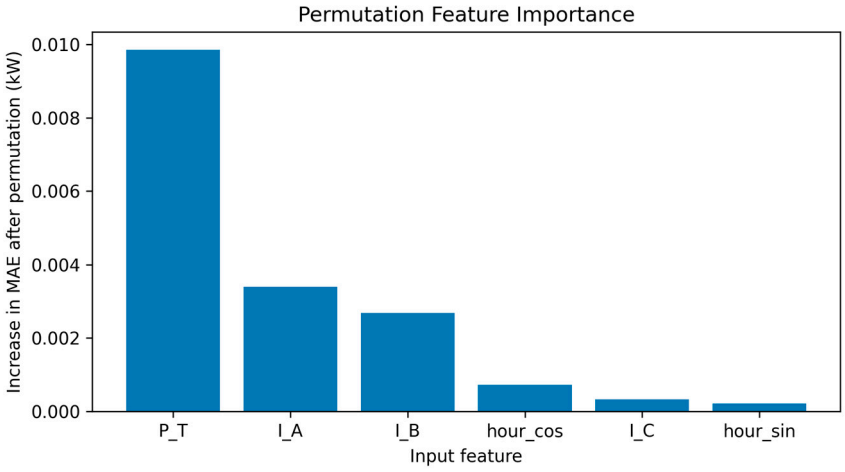


Figure 12. Permutation feature importance expressed as the increase in test MAE.

Table 10. Permutation feature importance for the fixed seed $\tau = 0.90$ model.

Rank	Feature	Base MAE (kW)	MAE After Permutation (kW)	ΔMAE (kW)
1	P_T	0.033423	0.043276	+0.009853
2	I_A	0.033423	0.036817	+0.003394

Table 10. *Cont.*

Rank	Feature	Base MAE (kW)	MAE After Permutation (kW)	ΔMAE (kW)
3	I_B	0.033423	0.036108	+0.002685
4	$hour_{cos}$	0.033423	0.034145	+0.000722
5	I_C	0.033423	0.033744	+0.000321
6	$hour_{sin}$	0.033423	0.033637	+0.000214

4.8. Computational and Deployment Characteristics

These measurements indicate that the compact model is computationally modest on a GPU. They do not establish latency on a CPU, embedded controller, or production metering system. Deployment at building or grid scale would also require reliable acquisition of synchronized three-phase currents, communication delay analysis, drift monitoring, and site-specific recalibration. The measured computational characteristics are summarized in Table 11.

Table 11. Proposed model computational characteristics on the experimental T4 runtime.

Measure	Value
Trainable parameters	46,914
Training time	46.46 ± 2.53 s per run
Epochs completed	21.2 ± 3.5
Inference latency	0.494 ± 0.166 ms per sample
Input requirement	20 real observations × 6 variables
Output	One upper quantile next observation forecast

5. Discussion

The final benchmark comparison changes the central interpretation of the work. GRU and CNN–LSTM are better point forecasters on this dataset, while the proposed Transformer is better aligned with an asymmetric upper-bound objective. Its value is not a universal reduction in MAE or RMSE; it is the substantial reduction in underprediction, particularly during peak targets, while maintaining approximately calibrated overall coverage.

The ablation clarifies the source of this behavior. Multivariate electrical inputs provide a modest deterministic improvement, especially through Phase A and Phase B currents. Pinball loss produces the much larger directional shift from negative to positive peak mean error. Additional depth raises coverage but does not improve point accuracy, indicating that loss design is more influential than model capacity for the present safety objective.

The quantile comparison shows why a single operating point should not be described as universally optimal. Lower τ values favor average accuracy and permit more violations; higher τ values provide stronger protection but increase conservative error. Selecting τ therefore depends on reserve cost, service criticality, and tolerance for false alarms. The $\tau = 0.90$ setting represents a middle operating point for this dataset.

The chaos horizon perspective discussed by Stergiou and Karakasidis [32] reinforces the need to connect claims to the actual forecast horizon. The present task predicts only the next native observation and does not claim prediction beyond a Lyapunov horizon. Similarly, the corrected timestamp audit removes the earlier assertion that resampling native 10-min data recovers one-to-three-minute physical transients.

From an applied perspective, the proposed model is best viewed as a complementary risk-aware layer. A deterministic model such as GRU may support routine scheduling where central accuracy is paramount, while a calibrated upper quantile forecast can support conservative alarms, reserve planning, or operator review. Combining both outputs may be more useful than replacing one objective with the other.

6. Limitations and Deployment Considerations

The study uses one low-voltage university laboratory. The results may not transfer to residential, commercial, feeder-level, or transmission-scale loads without external validation.

The source instrumentation records at approximately 10-min intervals. The analysis cannot establish one-to-three-minute current-inrush lead times or recover unobserved transients between measurements.

The task is next observation forecasting with a single historical window. Multi-step forecasts and explicit horizon-dependent calibration were not evaluated.

Weather, occupancy, equipment state labels, holidays, and calendar events were unavailable. Their inclusion may improve conditional calibration.

The $\tau = 0.90$ model is approximately calibrated overall but undercovers the peak subset. Its $73.36 \pm 5.32\%$ peak coverage is not a formal safety guarantee.

The proposed model was repeated across five seeds, whereas the benchmark neural models and $\tau = 0.75/0.95$ sensitivity runs used one fixed seed. Repeated runs for every comparator would strengthen uncertainty estimates.

The fixed-order SARIMA model is a computationally stable reference rather than an exhaustive seasonal-order search.

GPU latency does not establish real deployment latency. Edge hardware, communication delays, data quality alarms, and periodic recalibration require separate validation.

7. Conclusions

This study evaluated a multivariate quantile Transformer for safety-oriented next observation load forecasting in a university laboratory. A leakage-free timestamp audit and native observation pipeline replaced the earlier unsupported 3 min resampling interpretation. On the same test period, GRU achieved the best deterministic accuracy, while the proposed $\tau = 0.90$ Transformer achieved $87.95 \pm 1.01\%$ empirical coverage and reduced peak underprediction to $26.64 \pm 5.32\%$, compared with 72.94–100% for the conventional benchmark outputs. Multi-quantile experiments confirmed that greater safety margins come at the cost of larger point error. The ablation showed that multivariate currents provide a modest accuracy improvement and that pinball loss is the principal driver of peak-bias correction. These findings support the model as a complementary risk-aware forecasting layer for this facility-scale case study, not as proof of universal smart grid superiority. External validation, repeated benchmark runs, multi-step forecasting, and deployment testing are required before broader operational claims can be made.

Author Contributions: Conceptualization, M.O.A.E. and A.M.G.; methodology, M.O.A.E. and A.M.G.; software, M.H.E.; validation, M.O.A.E. and A.M.G.; formal analysis, M.O.A.E. and A.M.G.; investigation, M.H.E.; resources, M.O.A.E.; data curation, M.O.A.E. and M.H.E.; writing—original draft preparation, M.O.A.E. and M.H.E.; writing—review and editing, M.O.A.E., M.H.E. and A.M.G.; visualization, M.H.E. and A.M.G.; supervision, M.O.A.E.; project administration, M.O.A.E.; funding acquisition, M.O.A.E. and M.H.E. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The dataset analyzed in this study is publicly available in the Zenodo repository at https://doi.org/10.5281/zenodo.17107426 (accessed on 10 May 2026) and is described in the dataset publication cited in this article. The analysis code is available from the corresponding author upon reasonable request.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Akhtar, S.; Shahzad, S.; Zaheer, A.; Ullah, H.S.; Kilic, H.; Gono, R.; Jasiński, M.; Leonowicz, Z. Short-Term Load Forecasting Models: A Review of Challenges, Progress, and the Road Ahead. *Energies* **2023**, *16*, 4060. [CrossRef]
2. Abdulameer, Y.H.; Ibrahim, A.A. Forecasting of Electrical Energy Consumption Using Hybrid Models of GRU, CNN, LSTM, and ML Regressors. *J. Wirel. Mob. Netw. Ubiquitous Comput. Dependable Appl.* **2025**, *16*, 560–575. [CrossRef]
3. Wei, N.; Yin, C.; Yin, L.; Tan, J.; Liu, J.; Wang, S.; Qiao, W.; Zeng, F. Short-Term Load Forecasting Based on WM Algorithm and Transfer Learning Model. *Appl. Energy* **2024**, *353*, 122087. [CrossRef]
4. Adham, M.; Keene, S.; Bass, R.B. Distributed Energy Resources: A Systematic Literature Review. *Energy Rep.* **2025**, *13*, 1980–1999. [CrossRef]
5. Kolambe, M.; Arora, S. Forecasting the Future: A Comprehensive Review of Time Series Prediction Techniques. *J. Electr. Syst.* **2024**, *20*, 575–586. [CrossRef]
6. Chen, W.; Han, G.; Zhu, H.; Liao, L. Short-Term Load Forecasting with an Ensemble Model Based on 1D-UCNN and Bi-LSTM. *Electronics* **2022**, *11*, 3242. [CrossRef]
7. Le, T.; Vo, M.T.; Vo, B.; Hwang, E.; Rho, S.; Baik, S.W. Improving Electric Energy Consumption Prediction Using CNN and Bi-LSTM. *Appl. Sci.* **2019**, *9*, 4237. [CrossRef]
8. Engelbrecht, S.E.; Berthier, N.E.; O'Sullivan, L.P. The Undershoot Bias: Learning to Act Optimally under Uncertainty. *Psychol. Sci.* **2003**, *14*, 257–261. [PubMed]
9. Foumani, N.M.; Miller, L.; Tan, C.W.; Webb, G.I.; Forestier, G.; Salehi, M. Deep Learning for Time Series Classification and Extrinsic Regression: A Current Survey. *ACM Comput. Surv.* **2024**, *56*, 217. [CrossRef]
10. Fan, S.; Hyndman, R.J. Short-Term Load Forecasting Based on a Semi-Parametric Additive Model. *IEEE Trans. Power Syst.* **2012**, *27*, 134–141. [CrossRef]
11. Mahajan, H. Comparative Study of Arima, Lstm and Prophet Models for Time Series Forecasting: A Comprehensive Review. *Preprints* **2026**, 2026010264. [CrossRef]
12. Niu, Y.; Jia, X.; Lee, C.K.; Jiang, H.; Leh, C.P. Analyzing and Forecasting Laboratory Energy Consumption Patterns Using Autoregressive Integrated Moving Average Models. *Laboratories* **2025**, *2*, 2. [CrossRef]
13. Gomes, T.E.d.O.; Garcia, V.J.; Borniatti, A.R. An ARIMA Model-Based Approach to Improve Electricity Reliability. *Rev. Gestão Soc. E Ambient.* **2024**, *18*, e09855. [CrossRef]
14. Alberg, D.; Last, M. Short-Term Load Forecasting in Smart Meters with Sliding Window-Based ARIMA Algorithms. *Vietnam J. Comput. Sci.* **2018**, *5*, 241–249. [CrossRef]
15. Lakshmanan, P. Design of a Dynamic Demand Response Model through Intelligent Clustering Algorithm Based on Load Forecasting in Smart Grid. *Elektron. Elektrotech.* **2022**, *28*, 15–23. [CrossRef]
16. López, J.C.; Rider, M.J.; Wu, Q. Parsimonious Short-Term Load Forecasting for Optimal Operation Planning of Electrical Distribution Systems. *IEEE Trans. Power Syst.* **2018**, *34*, 1427–1437. [CrossRef]
17. Habiba, M.; Pearlmutter, B.A.; Maleki, M. Recent Trends in Modelling the Continuous Time Series Using Deep Learning: A Survey. *arXiv* **2024**, arXiv:2409.09106.
18. Prakash, A.; Singh, S.K. CO_2 Emissions Prediction from Coal Used in Power Plants Using Univariate and Multivariate Machine Learning Models. *Iran J. Comput. Sci.* **2024**, *7*, 533–549. [CrossRef]
19. Yang, L.; Shami, A. IoT Data Analytics in Dynamic Environments: From an Automated Machine Learning Perspective. *Eng. Appl. Artif. Intell.* **2022**, *116*, 105366. [CrossRef]
20. Chien, H.-Y.S.; Turek, J.S.; Beckage, N.; Vo, V.A.; Honey, C.J.; Willke, T.L. Slower Is Better: Revisiting the Forgetting Mechanism in LSTM for Slower Information Decay. *arXiv* **2021**, arXiv:2105.05944.
21. Mahjoub, S.; Chrifi-Alaoui, L.; Marhic, B.; Delahoche, L. Predicting Energy Consumption Using LSTM, Multi-Layer GRU and Drop-GRU Neural Networks. *Sensors* **2022**, *22*, 4062. [CrossRef] [PubMed]
22. Alsamraee, S.A.; Khanna, S. A Hybrid Transformer–TCN–GRU Based Model for Thermal Load Forecasting of a Large University Campus. *Energy* **2026**, *344*, 140114. [CrossRef]
23. Moveh, S.; Merchán-Cruz, E.A.; Abuhussain, M.; Dodo, Y.A.; Alhumaid, S.; Alhamami, A.H. Deep Learning Framework Using Transformer Networks for Multi-Building Energy Consumption Prediction in Smart Cities. *Energies* **2025**, *18*, 1468. [CrossRef]

24. Shwetha, B.N.; Harish Kumar, K.S. Prediction of Electricity Consumption in Residential Areas Using Temporal Fusion Transformer and Convolutional Neural Network. *J. Mach. Comput.* **2025**, *5*, 209–219. [CrossRef]
25. Liu, Y.; Zheng, R.; Liu, M.; Zhu, J.; Zhao, X.; Zhang, M. Short-Term Load Forecasting Model Based on Time Series Clustering and Transformer in Smart Grid. *Electronics* **2025**, *14*, 230. [CrossRef]
26. Pentsos, V.; Tragoudas, S.; Wibbenmeyer, J.; Khdeer, N. A Hybrid LSTM-Transformer Model for Power Load Forecasting. *IEEE Trans. Smart Grid* **2025**, *16*, 2624–2634. [CrossRef]
27. Xu, C.; Chen, G. Interpretable Transformer-Based Model for Probabilistic Short-Term Forecasting of Residential Net Load. *Int. J. Electr. Power Energy Syst.* **2024**, *155*, 109515. [CrossRef]
28. Aoufi, S.; Derhab, A.; Guerroumi, M.; Guemmouma, H.; Lazali, H. LITE-FORT: Lightweight Three-Stage Energy Theft Detection Based on Time Series Forecasting of Consumption Patterns. *Electr. Power Syst. Res.* **2023**, *225*, 109840. [CrossRef]
29. Yang, N.; Zhang, Y.; Wang, Y.; Tang, D.; Li, Y.; Yuan, D. Adaptformer: An Adaptive Multimodal Deep Decomposition Approach for Power Consumption Forecasting. In *Advanced Data Mining and Applications (ADMA 2024)*; Lecture Notes in Computer Science; Springer: Singapore, 2025; Volume 15391, pp. 48–62. [CrossRef]
30. Jadon, A.; Patil, A.; Jadon, S. A Comprehensive Survey of Regression-Based Loss Functions for Time Series Forecasting. In *Data Management, Analytics and Innovation*; Sharma, N., Goje, A.C., Chakrabarti, A., Bruckstein, A.M., Eds.; Lecture Notes in Networks and Systems; Springer: Singapore, 2024; Volume 998. [CrossRef]
31. Nygård, H.S.; Grøtan, S.; Kvisberg, K.R.; Rydin Gorjão, L.; Martinsen, T. Enhancing Peak Electricity Demand Forecasting for Commercial Buildings Using Novel LSTM Loss Functions. *Electr. Power Syst. Res.* **2025**, *246*, 111722. [CrossRef]
32. Stergiou, K.; Karakasidis, T.E. Application of Deep Learning and Chaos Theory for Load Forecasting in Greece. *Neural Comput. Appl.* **2021**, *33*, 16713–16731. [CrossRef]
33. Saldarriaga-Zuluaga, S.D.; Velasco-Méndez, J.R.; Moreno-Paniagua, C.M.; Alvarez-Arboleda, B.; Estrada-Mesa, S.A. Electrical Measurement Dataset from a University Laboratory for Smart Energy Applications. *Data* **2025**, *10*, 170. [CrossRef]
34. Mohassel, R.R.; Fung, A.S.; Mohammadi, F.; Raahemifar, K. A Survey on Advanced Metering Infrastructure and Its Application in Smart Grids. In Proceedings of the 2014 IEEE 27th Canadian Conference on Electrical and Computer Engineering (CCECE), Toronto, ON, Canada, 4–7 May 2014; pp. 1–8. [CrossRef]
35. Shao, J.; Zhong, B. Last Observation Carry-Forward and Last Observation Analysis. *Stat. Med.* **2003**, *22*, 2429–2441. [CrossRef] [PubMed]
36. Blackburne, B.P.; Whelan, S. Measuring the Distance between Multiple Sequence Alignments. *Bioinformatics* **2012**, *28*, 495–502. [CrossRef] [PubMed]
37. Elloumi, Y.; Khazem, S.; Krayem, I.; Mahesananthan, J. Cyclical Temporal Encoding for Ensemble Deep Learning in Multistep Energy Forecasting. *HAL Prepr.* **2025**, hal-05170016.
38. Kim, Y.-S.; Kim, M.K.; Fu, N.; Liu, J.; Wang, J.; Srebric, J. Investigating the Impact of Data Normalization Methods on Predicting Electricity Consumption in a Building Using Different Artificial Neural Network Models. *Sustain. Cities Soc.* **2025**, *118*, 105570. [CrossRef]
39. Zhang, J.; He, T.; Sra, S.; Jadbabaie, A. Why Gradient Clipping Accelerates Training: A Theoretical Justification for Adaptivity. In Proceedings of the International Conference on Learning Representations (ICLR), Virtual, 26 April–1 May 2020. Available online: https://openreview.net/forum?id=BJgnXpVYwS (accessed on 8 August 2026).

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.